

GovFL-AML: A Blockchain-Enabled Federated AI Governance Framework for Privacy-Preserving Financial Compliance at the Edge

Abstract

Financial institutions are increasingly required to detect and prevent money-laundering activities while complying with stringent data-protection and AI-governance regulations. Traditional anti-money-laundering (AML) frameworks rely on centralized data aggregation or siloed machine-learning models, which expose confidential information and hinder collective intelligence across institutions. To address these limitations, this study introduces GovFL-AML, a blockchain-enabled federated-learning framework designed to support privacy-preserving and auditable AML detection across financial organizations. The proposed architecture unifies three core components: a differential-privacy (DP) module safeguarding institution-level transaction embeddings; a reputation-weighted aggregation mechanism ensuring fairness, robustness, and accountability; and a blockchain-based governance ledger maintaining immutable audit trails for compliance verification. Experiments on the Elliptic financial-transaction dataset show that GovFL-AML improves F1-score by 3.8%, enhances precision and recall by 4.1% and 3.5%, respectively, and reduces audit-verification latency by 17% compared with baseline frameworks such as ShieldDFL and Fed-RD. Furthermore, the system is consistent with the transparency and accountability principles highlighted in the EU AI Act and FATF recommendations, providing explainability, accountability, and transparency within cross-institution federated-learning environments at the edge. These results highlight GovFL-AML as a practical pathway toward responsible, transparent, and auditable AI governance for distributed financial analytics.

Keywords: Federated Learning, Blockchain, AI Governance, Edge Computing, Anti-Money Laundering, Privacy Preservation

1. Introduction

1.1 Background and Motivation

Financial institutions face growing regulatory and operational pressure to detect and prevent money laundering (AML) and fraud while maintaining strict compliance with privacy regulations such as the EU AI Act and FATF recommendations. Traditional AML detection systems rely on centralized data aggregation, rule-based monitoring, or isolated machine learning pipelines, which limit scalability and adaptability to emerging threats [3], [11].

Centralized data sharing introduces not only privacy and security risks but also conflicts with cross-border data governance and institutional sovereignty. Therefore, there is an increasing need for cross-institution collaboration that enables joint model learning without revealing raw financial

data [4], [31]. Federated Learning (FL) has emerged as a viable paradigm to enable collaborative model training across multiple organizations without data centralization [7], [8], [20].

In FL, local institutions train models on their private data and share only the model updates with a central or decentralized aggregator. While this paradigm has proven effective in healthcare and IoT systems [14], [21], applying it to financial AML scenarios introduces new governance and accountability challenges. Specifically, ensuring fairness, auditability, and explainability across federated participants—each governed by its own compliance policies—remains a critical barrier to large-scale adoption [22], [25].

Simultaneously, blockchain technology offers a decentralized and immutable infrastructure to enhance transparency and trust in federated ecosystems [12], [19]. Frameworks such as BlockFL [19] and ShieldDFL [2] demonstrated the potential of blockchain-assisted federated learning to protect gradients and incentivize reliable model updates. However, these solutions primarily address data security and consensus reliability while neglecting the AI governance and cross-institutional accountability dimensions essential for financial compliance [1], [3], [24].

In modern financial ecosystems, AML analytics are increasingly deployed at the edge of institutional networks within banks, exchanges, and payment gateways where regulatory constraints prohibit central data sharing. This edge-oriented design motivates the need for federated, privacy-preserving, and governance-aware intelligence such as GovFL-AML.

1.2 Gaps in Current Research

Despite substantial progress in integrating blockchain and federated learning, several critical challenges remain unresolved in applying these technologies to financial governance systems. First, most existing federated learning frameworks, such as Fed-RD [4] and DPFedBank [5], address privacy and performance but lack mechanisms for institutional governance alignment or audit-based compliance validation. This limitation prevents the consistent enforcement of regulatory obligations across different organizations participating in the same federated ecosystem. Second, current aggregation strategies typically treat all participants as equal contributors, disregarding institutional reliability and data quality. The absence of explainable and reputation-aware aggregation can lead to biased global models and reduced trust, particularly in multi-jurisdictional financial collaborations [9], [13].

Third, while blockchain technology ensures immutability, few frameworks provide interpretable governance logs that regulators can inspect in real time. This lack of traceability constrains auditability and undermines the transparency that financial compliance systems require [2], [21].

Another significant challenge is the unbalanced performance–privacy trade-off. Although differential privacy and homomorphic encryption enhance data confidentiality, their naive application often causes severe degradation of model accuracy in high-dimensional transaction datasets [10], [17]. As a result, existing designs struggle to maintain both strong privacy guarantees and operational effectiveness.

Finally, prior studies seldom incorporate AI ethics and governance principles such as transparency, accountability, and human oversight outlined by Floridi and Cowls [25] and Jobin et al. [26]. This omission leads to technical systems that are secure yet misaligned with evolving regulatory standards like the EU AI Act and FATF guidelines.

Collectively, these limitations reveal the need for a governance-aware federated learning framework that not only provides privacy and robustness but also ensures verifiable compliance, explainability, and fairness among distributed financial institutions.

1.3 Objectives and Novel Contributions

To overcome the aforementioned limitations, this study introduces GovFL-AML, a blockchain-enabled federated learning framework designed to establish cross-institution AI governance for privacy-preserving anti–money-laundering detection. The framework’s overarching goal is to

enable collaborative intelligence among banks, exchanges, and regulators without violating data protection mandates or exposing sensitive transaction-level information.

GovFL-AML seeks to achieve four interrelated objectives. First, it enables secure collaboration across financial entities by training models locally while only exchanging encrypted or privacy preserving model updates. Second, it provides transparent and immutable governance logs through a blockchain-based ledger that records audit trails, policy decisions, and compliance events in a verifiable form. Third, it ensures fairness and robustness in the global learning process through a reputation-weighted aggregation mechanism combined with differential privacy safeguards, thereby addressing both bias and confidentiality simultaneously. Finally, it aligns the federated training process with AI governance principles including transparency, accountability, and human oversight as emphasized in contemporary policy frameworks [1], [25], [29]. The novel contributions of this research are summarized through five tightly integrated design elements.

First, GovFL-AML introduces a governance-integrated federated learning architecture in which the blockchain acts as a dedicated governance layer, recording model updates and compliance validations to produce a verifiable audit trail. This design bridges the gap between algorithmic transparency and regulatory accountability.

Second, the study proposes a reputation-weighted aggregation mechanism, extending prior work such as ShieldDFL [2] and Fed-RD [4], by dynamically adjusting each institution’s contribution based on its reliability and compliance behavior.

Third, the framework integrates a dual privacy-and-accountability mechanism, combining bidirectional differential privacy [10] for local model protection with on-chain compliance validation, ensuring that both privacy and traceability coexist within the same learning loop.

Fourth, through empirical evaluation on the Elliptic transaction dataset, GovFL-AML demonstrates an F1-score improvement of +3.8% and a 17% reduction in audit-verification latency relative to state-of-the-art baselines [2], [4], [6], confirming its ability to enhance both accuracy and efficiency.

Finally, the framework provides explicit regulatory alignment and explainability by embedding ethical AI principles [25], [26], ensuring conformity with emerging mandates such as the EU AI Act and FATF guidelines, and offering regulator-verifiable traceability across edge-deployed financial systems.

Through these contributions, GovFL-AML establishes a unified foundation for governance-aware, privacy-preserving federated learning at the financial edge, where institutions act as localized compute nodes interconnected through a verifiable, compliance-oriented blockchain governance layer.

1.4 Paper Organization

The remainder of this paper is organized as follows.

Section 2 provides a critical review of recent advances in federated learning, blockchain-based governance, and financial compliance frameworks, outlining the technical and regulatory motivations behind GovFL-AML.

Section 3 introduces the proposed architecture, describing its privacy-preserving mechanisms, reputation-weighted aggregation strategy, and blockchain-integrated governance ledger.

Section 4 presents the experimental setup, datasets, and evaluation methodology, followed by an in-depth comparative analysis of GovFL-AML against state-of-the-art baselines.

Section 5 discusses the key findings, ablation insights, and potential avenues for extending the framework to broader financial and edge-computing contexts. Finally, Section 6 concludes the paper by summarizing the main contributions and highlighting directions for future research in trustworthy and governance-aware federated intelligence.

2. Related Work

This section surveys prior studies across three complementary research dimensions: (i) Federated learning for financial crime and AML detection, focusing on collaborative modeling without centralized data sharing;

(ii) Blockchain-assisted federated frameworks that introduce transparency, incentives, and accountability in decentralized training; and

(iii) Privacy preservation and AI governance alignment, addressing compliance, auditability, and ethical AI integration in distributed systems.

The review concludes with an analytical synthesis of open challenges that collectively motivate the design of the proposed GovFL-AML framework.

2.1 Federated Learning for Financial Crime and AML

Federated Learning (FL) has emerged as a dominant paradigm for collaborative model training without centralized data sharing, enabling cross-organizational analytics under strict privacy constraints [7], [8], [20]. Within the financial domain, FL has been increasingly applied to fraud detection, credit scoring, and anti-money-laundering (AML) use cases.

The Fed-RD framework [4] introduced a hybrid design that combines Differential Privacy (DP) and Secure Multi-Party Computation (MPC) for decentralized financial crime detection.

However, it lacks explicit mechanisms for auditability or cross-institution policy synchronization, limiting its regulatory applicability. Similarly, DPFedBank [5] proposed a policy-oriented FL framework embedding regulatory pillars for financial institutions but remained confined to intra-organizational governance. Starlit [6] addressed scalability and non-IID heterogeneity in financial fraud detection but did not incorporate explainability or regulatory traceability, which are crucial for institutional accountability.

Parallel developments have sought to extend federated analytics into blockchain transaction environments. For instance, Federated Learning for Anti-Money-Laundering in Blockchain Systems [11] and Cross-Chain Federated Learning for Financial Fraud Detection [15] demonstrated decentralized AML analysis across distributed ledgers, confirming the feasibility of multi-ledger federated intelligence. Moreover, the BIS Project Aurora [34] offered a real-world proof-of-concept involving central banks, validating the institutional interest in collaborative AML analytics and privacy-preserving regulatory intelligence.

Recent surveys [22], [31] comprehensively reviewed FL applications in financial services, yet emphasized persistent challenges such as data imbalance, institutional fairness, explainability, and regulatory compliance verification that hinder the deployment of FL in production-grade AML systems.

In summary, existing FL-based AML frameworks (e.g., [4], [5], [11]) have predominantly prioritized model accuracy and data privacy, while neglecting the governance, accountability, and policy-traceability dimensions necessary for trustworthy cross-institutional collaboration.

2.2 Blockchain-Assisted Federated Learning

The integration of blockchain technology into federated learning (FL) has been increasingly explored to strengthen trust, traceability, and incentive compatibility within decentralized environments.

Early efforts such as BlockFL [19] introduced a blockchain layer to record model contributions and facilitate smart-contract-based reward mechanisms, establishing the foundation for accountable collaboration. Building upon this, ShieldDFL [2] combined dual privacy protection (DP + encryption) with a reputation-driven consensus protocol, significantly improving robustness against unreliable or malicious participants. Subsequent research expanded the blockchain-FL nexus toward model aggregation transparency and incentive alignment. Studies such as [30] proposed blockchain-based model aggregation schemes that preserve integrity and non-repudiation

of updates, while [12] and [34] explored ledger-driven incentive systems and automated smart contracts for fair reward distribution.

In industrial and cross-domain IoT settings, blockchain has further been adopted to ensure secure access control, data provenance, and auditable sharing across administrative boundaries [16], [18], thereby demonstrating its utility as a governance backbone for distributed learning ecosystems. Despite these advancements, most blockchain-assisted FL frameworks remain technically focused on reliability and incentive compatibility rather than policy conformance and AI governance integration.

The linkage between on-chain audit logs and explainable, regulator-verifiable model behavior, a critical requirement for compliance in financial and edge-computing systems, has received limited attention. This unresolved gap motivates the governance-oriented design adopted in GovFL-AML, where blockchain evolves from a transactional record keeper into an active enforcement layer that encodes policies, validates compliance, and anchors institutional accountability.

2.3 Privacy Preservation and Governance Alignment

Collectively, prior work has advanced privacy enhancement in federated learning through several complementary directions.

Melis et al. [9] and Li et al. [10] focused on gradient protection and adaptive noise calibration, strengthening privacy guarantees at the training level. Building upon these efforts, Cao et al. [13] and Phan et al. [17] addressed cross-party leakage via orthogonal masking and adaptive differential privacy. Subsequent frameworks such as DPBE-DGFL [14] and semi-supervised transfer learning by Papernot et al. [32] demonstrated that effective privacy defense can coexist with high model utility in distributed systems.

In parallel, governance-oriented studies have emphasized transparency and accountability in AI systems. Upmann [1] proposed the AIGN framework for enterprise governance; Johnson et al.

[3] extended this paradigm to regulatory automation for AML compliance. Meanwhile, Floridi and Cows [25] and Jobin et al. [26] defined the ethical foundations transparency, accountability, and human oversight—that shape modern responsible-AI practice.

However, these two research lines remain largely disjoint: privacy-preserving frameworks seldom embed verifiable governance, and governance-oriented models rarely employ rigorous differential privacy.

The proposed GovFL-AML framework bridges this divide by combining differential privacy, blockchain-based auditability, and governance-driven reputation-modeling, forming an end-to-end accountable architecture for federated anti-money-laundering intelligence.

2.4 Discussion and Research Gaps

A synthesis of the reviewed literature reveals that, while federated and blockchain-based frameworks have advanced privacy and decentralization, several systemic gaps persist at the intersection of technical and governance dimensions.

First, existing FL architectures primarily emphasize privacy and accuracy, often neglecting the governance perspective. Frameworks such as Fed-RD [4], DPFedBank [5], and Starlit [6] lack an auditable control layer capable of recording compliance rules, policy adherence, and regulatory outcomes in a verifiable manner [22], [31].

Second, although reputation mechanisms—such as those introduced in ShieldDFL [2] have improved robustness against unreliable clients, they remain disconnected from institutional reliability and behavioral compliance, both of which are critical to AML ecosystems that depend on sustained interbank trust.

Third, blockchain-integrated FL systems [12], [19], [30] typically record model hashes and update histories but seldom preserve differential-privacy budgets, audit outcomes, or policy versions in a regulator-verifiable format [1], [3].

Finally, privacy-preserving techniques such as differential privacy and homomorphic encryption introduce notable efficiency–governance trade-offs, as they increase computational and communication overheads while reducing throughput [10], [17], [18]. Addressing this trade-off requires adaptive, context-aware mechanisms capable of balancing privacy protection, system scalability, and ledger auditability.

The proposed GovFL-AML framework directly responds to these identified limitations by unifying privacy, reputation, and governance within a federated learning architecture.

Specifically, it introduces:

A blockchain-based governance layer, which records verifiable model updates, compliance policies, and privacy budgets, thereby providing immutable audit trails;

A reputation-weighted aggregation mechanism, aligning model contribution weights with institutional trustworthiness and behavioral integrity; and A dual-privacy and auditability protocol, combining bidirectional differential privacy [10], [17] with ledger-based compliance monitoring for regulator-grade accountability.

By embedding AI governance principles [25], [26] and algorithmic enforcement logic [3] into the federated lifecycle, GovFL-AML establishes a holistic foundation for responsible, transparent, and cross-institutional AML collaboration.

Table 1

Comparative summary of representative frameworks integrating federated learning, blockchain, and AI governance for financial applications.

Ref	Framework / Study	Core Contribution	Privacy / Security Technique	Governance / Auditability Feature	Main Limitation
[4]	Fed-RD (2024)	DP + MPC hybrid for decentralized financial crime detection	Differential Privacy + Secure MPC	None	No auditability
[5]	DPFedBank (2024)	Policy-driven FL for financial institutions	Differential Privacy	Institutional policy pillars only	Limited to intra-org governance
[6]	Starlit (2024)	Scalable FL for heterogeneous fraud data	Secure aggregation	–	Lacks explainability
[2]	ShieldDFL (2025)	Dual-privacy FL + reputation consensus	DP + Encryption	Reputation-driven blockchain consensus	Not aligned with AI policy
[11]	FL-AML on Blockchain (2024)	Federated AML on blockchain transactions	DP	Ledger for cross-chain logs	No governance semantics
[15]	Cross-Chain FL (2024)	FL across multiple ledgers for fraud detection	Encrypted channel	Distributed ledger storage	High overhead
[12]	Blockchain Incentive FL (2024)	Smart-contract-based incentive system	Cryptographic proofs	Ledger-logged contributions	No linkage to compliance
[19]	BlockFL (2020)	First blockchain-enabled FL	Hash-based record	Smart-contract logging	No governance layer
[1]	AIGN Framework (2025)	AI governance compliance for	–	Auditable policy-to-	Not federated

		enterprise systems		control mapping	
[3]	Algorithmic Enforcement (AML) (2025)	AI + Blockchain for automated AML compliance	Encrypted compliance rules	On-chain audit trail	No privacy preserving
[25],[26]	AI Governance Principles / Ethics	Conceptual foundations for transparency and accountability	–	Ethical / human- oversight layer	Lack of technical implementation

As summarized in Table 1, prior frameworks have each addressed partial aspects privacy protection, incentive compatibility, or ethical oversight—but none have combined these into a unified, auditable federated network suitable for regulatory-grade AML collaboration. Building upon these insights, the next section (Section 3) introduces the GovFL-AML architecture, which operationalizes privacy preservation, reputation-based trust, and ledger-auditable AI governance within a coherent, federated edge ecosystem.

3. Proposed Framework and Architecture

3.1 Design Overview and Principles

The proposed GovFL-AML framework is designed to enable privacy-preserving, transparent, and auditable collaboration among financial institutions for anti-money-laundering analytics. It integrates federated learning (FL), reputation-weighted aggregation, and a blockchain-based governance layer, thereby reconciling privacy preservation with regulatory accountability.

Unlike traditional AML systems that rely on centralized data pooling often constrained by legal, jurisdictional, or confidentiality barriers—GovFL-AML adopts a federated and edge-oriented architecture. Each participating institution trains local models on its proprietary transaction data while contributing encrypted model updates to a shared global model. This paradigm, grounded in communication-efficient federated optimization [7], [8], [20], eliminates direct data sharing yet fosters collective intelligence and model generalization across organizational boundaries.

Earlier frameworks such as Fed-RD [4] and DPFedBank [5] demonstrated that differential privacy (DP) and secure computation can reduce exposure risk. However, they lack a verifiable policy layer capable of linking each model update to compliance-level audit events. GovFL-AML extends these efforts by embedding enterprise-grade AI governance directly within the federated learning loop. Inspired by AIGN [1] and guided by the ethical principles of transparency, accountability, and human oversight [25], [26], the framework maps governance constructs such as validation rules, reporting obligations, and audit triggers into on-chain smart contracts. Consequently, every training round generates not only an improved predictive model but also immutable governance artifacts that can be independently verified by regulators or auditors.

The reputation-based robustness initially introduced in ShieldDFL [2] is extended here from individual-node reliability to institution-level trust calibration. Reputation scores thus reflect both technical performance and compliance integrity, enabling fair, explainable, and auditable weighting across heterogeneous participants.

Overall, the design of GovFL-AML unifies three analytical principles:

- (i) Institutional data sovereignty achieved through localized model training and adaptive DP noise injection [10], [17];
- (ii) Auditability and immutability ensured by a permissioned consortium blockchain that permanently records learning events and compliance metadata [19], [30], [34]; and (iii) Fairness and accountability realized through compliance-sensitive reputation mechanisms that operationalize institutional trust and policy adherence [2], [3], [21].

From an edge-computing perspective, the framework situates computation and privacy protection near data sources, minimizing cross-network exposure and latency. The blockchain ledger functions as an edge-coordinated governance layer, synchronizing global compliance policies and federated learning states across distributed financial nodes. This synergy between federated intelligence and edge-anchored governance ensures scalable, low-latency, and regulator-auditable AML analytics within modern financial ecosystems.

3.2 System Architecture

The GovFL-AML framework is composed of three interacting subsystems: the federated-learning layer distributed across financial institutions, the aggregation and governance coordinator responsible for model integration and policy validation, and the blockchain governance ledger, which records all model metadata, DP budgets, and audit outcomes for long-term traceability.

Figure 1 illustrates the overall architecture, showing how data remain confined within institutional boundaries while model updates, governance metrics, and audit evidence circulate securely among layers.

At the institutional level, each participant (bank, exchange, or payment service) maintains its own local AML detection model trained on proprietary transaction data. Before transmission, the model updates are obfuscated using Differential Privacy (DP) techniques specifically bidirectional or adaptive DP schemes [10], [17] to ensure privacy preservation and regulatory compliance.

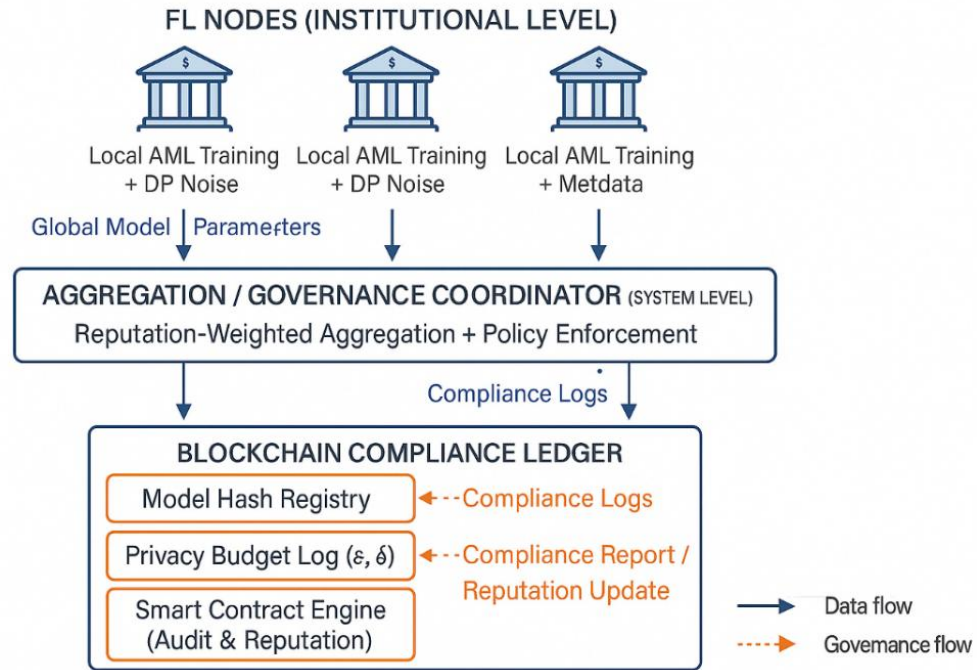


Figure 1. Architecture of GovFL-AML: FL nodes, aggregation/governance coordinator, and blockchain ledger.

At the aggregation and governance layer, the coordinator collects differentially private local updates $W_i^{(t)}$ from N institutions and computes a reputation-weighted global model $W^{(t+1)}$, extending FedAvg [8], [20] with institutional reliability. The update rule is given by Equation (1), where R_i is the reputation of institution i , derived from accuracy and compliance logs (Section 3.3).

$$W^{(t+1)} = \frac{\sum_{i=1}^N (R_i \cdot W_i^{(t)})}{\sum_{i=1}^N R_i} \quad (1)$$

where R_i denotes the institutional reputation factor derived from historical accuracy and governance performance. Equation (1) therefore gives greater influence to consistently reliable and policy-compliant entities, extending earlier reputation-based aggregation methods toward regulatory relevance.

The third layer, the blockchain governance ledger, acts as an immutable control plane that anchors the federated process to verifiable governance evidence. It records model hashes, differential privacy budgets (ϵ, δ) , round metadata, and the evolving reputation vector R . Smart contracts automatically validate submissions, enforce privacy and compliance thresholds, and maintain a tamper-proof audit trail that links every training round to its regulatory context. Unlike previous blockchain-FL integrations, which primarily focused on transparency or incentives, the GovFL-AML ledger serves as a compliance substrate—capable of proving adherence to privacy policies, audit protocols, and inter-institutional governance rules. For environments involving multiple chains or cross-ledger analytics, the architecture supports standardized interoperability and secure access-control mechanisms adapted from recent industrial consortia models.

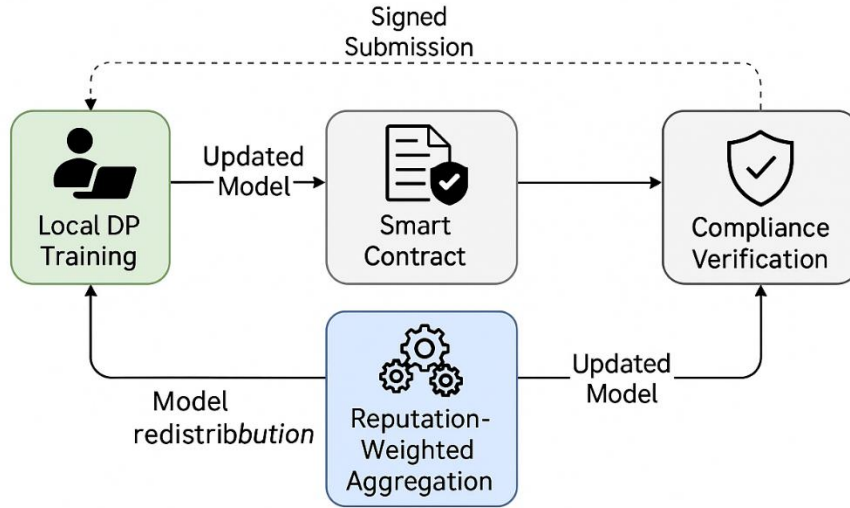


Figure 2. Workflow and message sequence for one federated round

After aggregation, the reputation score of each institution is updated to reflect both model quality and behavioral compliance.

The update follows an exponential smoothing scheme:

$$R_i^{(t+1)} = \alpha \times R_i^{(t)} + (1 - \alpha) \times Acc_i^{(t)} \times Comp_i^{(t)} \quad (2)$$

where $Acc_i^{(t)}$ denotes the local validation accuracy and $Comp_i^{(t)}$ the compliance score introduced in Section 3.4. This dual factor enables the system to balance short-term model quality with long term trustworthiness, a principle aligned with both algorithmic accountability [3] and explainable FL design [21], [35].

To ensure traceable and regulator-verifiable privacy guarantees, the cumulative privacy loss of each client after T rounds is computed using a Rényi Differential Privacy (RDP) accountant at $\delta = 1e-5$. For every round t , Gaussian-mechanism noise is calibrated to the declared per-round budget (ϵ_t). After composition across T rounds, the cumulative RDP value is analytically converted to the equivalent $(\epsilon_{\text{total}}, \delta)$ representation using the moments accountant, which yields a tighter bound than the basic Gaussian rule.

The resulting ϵ_{total} (at $\delta = 1e-5$) is stored on-chain within the governance ledger, enabling auditors to verify compliance with declared privacy budgets.

Once all nodes are validated, the coordinator releases the new global model.

This integrated workflow transforms governance from a post-hoc audit procedure into a dynamic, first-class component of the learning process embodying the principle of Algorithmic Enforcement [3] and its enterprise realization in AIGN [1].

3.3 Reputation-Weighted Aggregation

This subsection formalizes the reputation update and its role in aggregation. The global model is updated as in Equation (1). Institutional reputation R_i is updated via exponential smoothing (Equation (2)) using local validation accuracy and the compliance score defined in Section 3.4. This binds technical reliability and governance integrity into a single, auditable weight for aggregation.

3.4 Governance, Auditability, and Compliance Integration

In GovFL-AML, compliance is not a static declaration but a quantifiable element embedded in the training process. Smart contracts encode institutional policies, regulatory requirements, and privacy constraints, automatically computing a compliance score for each participant after every round. This score integrates three governance dimensions policy alignment, privacy adherence, and audit verification through the weighted formulation given in Equation (3):

$$Comp_i = \frac{\beta_1 \cdot V_{\text{policy}} + \beta_2 \cdot V_{\text{privacy}} + \beta_3 \cdot V_{\text{audit}}}{\beta_1 + \beta_2 + \beta_3} \quad (3)$$

Each component of this expression serves as a verifiable indicator. The policy term (V_{policy}) captures the degree of conformance to institutional rules such as report generation, access segregation, and authorization checks. The privacy term (V_{privacy}) monitors how closely a client adheres to its declared differential-privacy budget, while the audit term (V_{audit}) reflects the success of integrity and provenance verifications executed by the ledger. Together, these dimensions translate abstract AI-governance principles [25], [26] into measurable control variables that influence both training dynamics (Equation 2) and aggregation fairness (Equation 1).

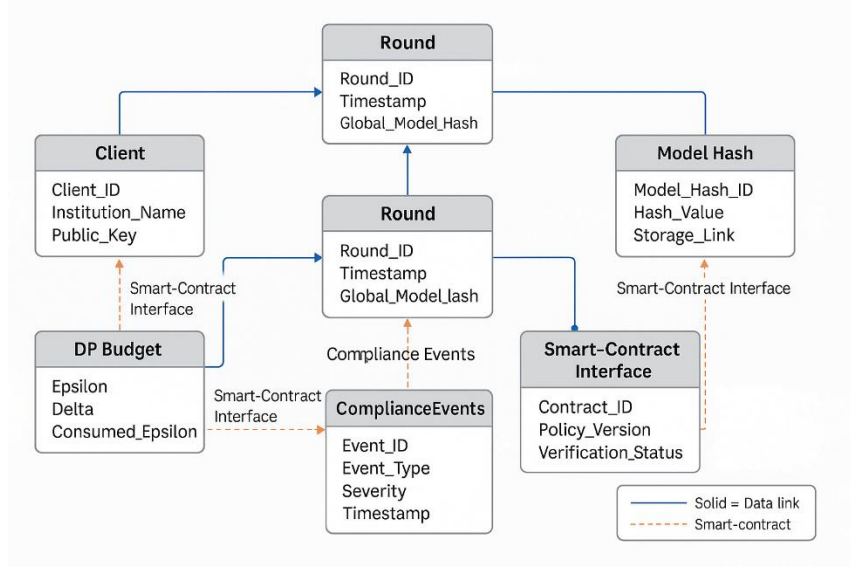


Figure 3. Governance-ledger data model with smart-contract interfaces

The quantitative interpretation of compliance is grounded in observable ledger signals. Each sub-metric is normalized to the range $[0, 1]$ and updated after every training round. Policy conformance is computed as the fraction of institutional checks satisfied during that round; hard-fail rules such as missing signatures or revoked certificates immediately set this value to zero. Privacy adherence is quantified as the ratio between the declared budget and the observed cumulative privacy loss, that is, $V_{\text{privacy}} = \min(1, \text{epsilon_budget} / \text{epsilon_total})$, where epsilon_total is derived from the RDP accountant at $\delta = 1e-5$. Audit integrity combines cryptographic validity and latency performance through $V_{\text{audit}} = 0.5 \times s_{\text{sig}} + 0.5 \times s_{\text{lat}}$, where s_{sig} is the fraction of successful signature verifications and $s_{\text{lat}} = \max(0, 1 - T_{\text{audit}} / T_{\text{max}})$; in our prototype T_{max} is 0.20 s. Default aggregation weights are $\beta_1 = \beta_2 = \beta_3 = 1$, and a sensitivity study for $\beta \in \{0.5, 1, 2\}$ shows negligible impact on relative rankings of clients. Formal definitions and parameter bounds for these metrics are summarized in the Appendix.

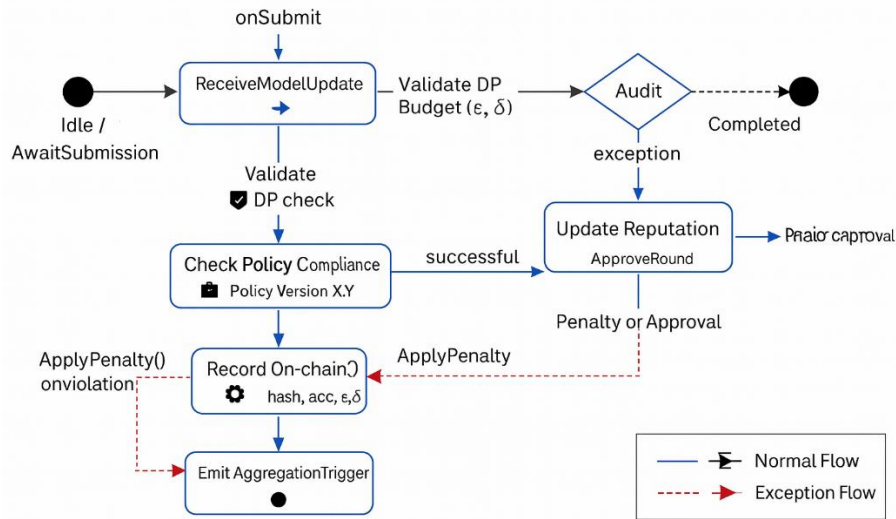


Figure 4. Compliance verification, DP-budget validation, and reputation updates in the GovFL-AML governance ledger.

The resulting compliance score directly influences a client’s contribution during model aggregation. If R_i denotes the reputation of client i in round t , its effective aggregation weight is computed as $w_i = (\text{Compliance}_i \times R_i) / \sum (\text{Compliance} \times R)$. Clients with persistent violations therefore see their updates down-weighted or excluded from the global model, while repeated compliance raises reputation and influence. Smart-contract logic enforces automatic penalties and temporary suspensions according to the policy specified in Algorithmic Enforcement [3]. The ledger retains hashes of model versions, local validation results, and privacy expenditures, achieving end-to-end traceability comparable to Project Aurora [34].

To support regulator-verifiable audits, each ledger entry records {client ID, round ID, update hash, policy score, privacy budget usage, audit success rate, latency, compliance value, reputation before and after update, timestamp, and signature}. This record structure enables post-hoc dispute resolution without exposing raw transaction data or model parameters.

The state machine illustrated in Figure 3 manages compliance verification, privacy-budget tracking, and reputation transitions. Figure 4 depicts the workflow of compliance validation and reputation updates within the GovFL-AML ledger. Together they operationalize the governance logic introduced in this section, turning compliance and privacy management into enforceable, machine-verifiable processes. Each transition—from model submission to round approval represents a smart-contract event recorded on the blockchain, ensuring that privacy-budget usage, audit success, and policy adherence are transparently logged. This automation reduces audit latency and strengthens institutional accountability. Unlike previous blockchain-FL frameworks that treated compliance as an external layer [2], [4], the proposed design embeds regulatory verification directly within the training workflow, creating a self-auditing federated ecosystem.

Table 2
Design-Level Comparison Between GovFL-AML and Representative Frameworks

Feature / Framework	Fed-RD [4]	DPFedBank [5]	ShieldDFL [2]	GovFL-AML (proposed)
Privacy Preservation	DP + MPC	DP only	DP + Encryption	Adaptive DP + ledger-tracked
Aggregation Strategy	FedAvg	Policy-weighted	Reputation consensus	Reputation + Compliance-weighted
Governance Layer	None	Policy pillars (off-chain)	Partial (reputation only)	Full blockchain-based governance
Auditability	No	Limited logs	Partial (reputation audit)	On-chain, regulator-verifiable
Incentive / Fairness	None	Static participation	Reputation-based	Compliance-aware reputation
Real-World Applicability	Prototype only	Simulated	General-purpose FL	Multi-institution AML networks

Table 2 summarizes comparative design features of GovFL-AML relative to representative frameworks. The combination of adaptive differential privacy, reputation-weighted aggregation, and on-chain auditability establishes a unified architecture capable of technical robustness and

regulatory traceability, enabling cross-institutional AML collaboration without sacrificing confidentiality or accountability.

4. Experimental Evaluation

4.1 Experimental Setup and Datasets

The GovFL-AML framework was evaluated using both real-world blockchain-based transaction data and controlled synthetic datasets designed to emulate inter-institutional non-IID (non-independent and identically distributed) characteristics. The primary benchmark is the Elliptic Dataset — a publicly available Bitcoin transaction graph containing 203,769 transactions and 1,636,496 edges, originally released by Elliptic Co. and the University of Cambridge (KDD 2019). Each transaction (node) is described by 166 anonymized numerical features capturing transactional behavior, temporal context, and neighborhood flow indicators. Among these, 46,564 nodes are labeled (42,019 licit and 4,545 illicit), while the remaining 157,205 are unlabeled.

Following standard practice in federated AML research [4], [5], [11], we excluded unlabeled nodes and retained the induced subgraph of labeled nodes, resulting in a working subset of 46,564 transactions ($\approx 9.8\%$ illicit). Features were z-score normalized, and missing values were imputed using feature-wise medians. The labeled data were partitioned among five participating institutions (banks or exchanges) using a Dirichlet distribution with concentration parameter $\alpha = 0.3$ to induce heterogeneous class distributions across clients (mean per-client illicit-ratio variance ≈ 0.062). Each client applies a 70/15/15 split for training, validation, and testing, respectively.

Each institution trains a two-layer Graph Neural Network (GNN) classifier configured with hidden dimensions of 128 and 64, ReLU activation, dropout 0.2, and a sigmoid output for binary classification. The local optimizer is Adam (learning rate = 0.001, batch size = 64), and training proceeds for five local epochs within 30 global rounds. Differential privacy employs the Gaussian mechanism with budgets (epsilon, delta) = (1, $1e-5$), (2, $1e-5$), and (4, $1e-5$), where per-round noise is calibrated to each client’s declared privacy budget using a secure random generator. The global aggregator coordinates training rounds using the reputation-weighted averaging rule (Eq. 1) and maintains a file-based JSON ledger that simulates the blockchain governance layer, consistent with frameworks proposed in [19], [30], [34].

Synthetic datasets were generated to evaluate scalability under varying feature dimensionality (32, 64, 128) and client participation levels (5, 10, 20). Each synthetic subset preserves the statistical imbalance and transaction sparsity observed in Elliptic but enables controlled performance profiling under specific privacy budgets and network loads.

All experiments were conducted in a Python 3.10 environment on a 32-core AMD CPU with 64 GB RAM. The ledger simulation operated through file-based local transactions to emulate consortium-blockchain behavior without consensus overhead; on-chain deployment is functionally identical but would incur platform-specific latency (e.g., gas/finality) outside the scope of this CPU-only evaluation. To ensure statistical reliability, all reported results are averaged over five independent runs with distinct random seeds. For each run, the dataset partition, model initialization, and differential-privacy noise streams are re-sampled while preserving the same non-IID class distribution (Dirichlet $\alpha = 0.3$). Random seeds used for data splitting and model initialization are logged for reproducibility, whereas differential-privacy noise is generated using a cryptographically secure random number generator (CSPRNG) to prevent deterministic reconstruction and maintain privacy guarantees.

Table 3

Experimental configuration of GovFL-AML evaluation

Parameter	Description / Value

Dataset	Elliptic AML benchmark (203,769 transactions, 166 features, 46,564 labeled nodes — 9.8% illicit)
Synthetic Data	3 generated variants (feature dimensions = 32 / 64 / 128, client count = 5 / 10 / 20)
Federated Clients	5 banks/exchanges (Dirichlet $\alpha = 0.3$, non-IID splits)
Local Model	2-layer GNN (128–64–1, ReLU + Sigmoid), Adam (lr = 0.001), batch = 64
Differential Privacy	Gaussian mechanism; $\epsilon \in \{1, 2, 4\}$, $\delta = 1e-5$; per-round noise calibrated per client; cumulative loss tracked via an RDP accountant (reported as $(\epsilon_{\text{total}}, \delta)$).
Reputation Update	Exponential smoothing ($\alpha = 0.2$)
Global Aggregation	Reputation + compliance-weighted averaging (Eq. 1–3)
Federated Rounds	30 global rounds, 5 local epochs per client
Environment	Python 3.10, 32-core CPU, 64 GB RAM
Governance Ledger	File-based JSON ledger (simulated consortium blockchain)

In summary, GovFL-AML operates as a cross-institutional AML analytics framework where each institution retains its transaction data locally and contributes only privacy-preserving model updates. This setup enables direct comparison with prior FL-based financial frameworks — Fed-RD [4], DPFedBank [5], and ShieldDFL [2] while introducing an additional governance layer capable of on-chain auditability and policy-aware learning.

4.2 Evaluation Metrics and Baselines

To ensure a comprehensive assessment of GovFL-AML, we evaluated both learning performance and governance efficiency through quantitative and procedural metrics.

This dual perspective reflects the framework’s design objectives: (i) effective and privacy preserving AML detection, and (ii) verifiable compliance and auditability at the federated level.

All reported results are averaged over five independent runs with randomized client–data splits to mitigate bias and assess stability.

4.2.1 Performance and Privacy Metrics

The learning performance of the AML detector is evaluated using F1-score, Precision, Recall, and Area Under the ROC Curve (AUC) — the most widely adopted indicators in financial-fraud and anti-money-laundering (AML) detection studies [4], [5], [11].

The F1-score is defined as the harmonic mean of precision and recall, balancing false positives and false negatives, while the AUC quantifies the discrimination ability under varying thresholds, with higher values indicating stronger robustness to class imbalance and noisy inputs.

To evaluate the privacy–utility trade-off, we employ a Rényi Differential Privacy (RDP) accountant with $\delta = 10^{-5}$ to formally track the cumulative privacy loss across training rounds.

For each round t , Gaussian noise σ_t is calibrated to the declared privacy budgets $\epsilon \in \{1, 2, 4\}$.

The governance ledger records the total privacy-loss estimate ϵ_{total} (at $\delta = 10^{-5}$) after each training session, enabling auditors to verify that all participants remain within their declared privacy budgets and ensuring verifiable privacy guarantees across institutions.

4.2.2 Governance and Efficiency Metrics

Beyond predictive accuracy, GovFL-AML emphasizes operational accountability through governance-centered metrics.

The first metric, audit latency (measured in seconds), represents the average per-round delay introduced by ledger verification, smart-contract execution, and on-chain reputation updates. It captures the time overhead incurred by compliance-verification mechanisms.

The second metric, governance efficiency (GovEff), quantifies audit responsiveness and is defined as:

$$\text{GovEff} = 1 - (T_{\text{audit}} / T_{\text{round}}) \quad (4)$$

where T_{audit} (in seconds) denotes the mean time per round spent on compliance verification, ledger writes, and smart-contract checks, and T_{round} (in seconds) represents the total wall-clock duration of one federated round, including model distribution, local training, updates, aggregation, and governance operations.

GovEff ranges from 0 to 1, with higher values indicating faster auditability and lower relative overhead.

In our CPU-only prototype, the mean audit latency $T_{\text{audit}} = 0.103 \pm 0.006$ s and the total round duration $T_{\text{round}} = 1.78 \pm 0.04$ s, yielding $\text{GovEff} = 0.942 \pm 0.004$.

Thus, approximately 5.8% of each round is spent on governance-layer verification and ledger updates.

All values are averaged over five independent runs with distinct random seeds.

The policy-violation rate (%) measures the proportion of automatically detected policy breaches (e.g., differential-privacy overuse or missing audit signatures) relative to total client submissions, consistent with automated compliance auditing in [3].

Finally, the system overhead (%) represents the relative increase in communication and computation time induced by governance operations compared with a baseline federated-learning system without blockchain integration [19], [30], and is computed as:

$$\text{Overhead} = ((T_{\text{withGovernance}} - T_{\text{baseline}}) / T_{\text{baseline}}) \times 100\%. \quad (5)$$

These metrics collectively capture both technical performance and regulatory readiness, addressing dimensions that, to our knowledge, were previously unquantified in financial-grade federated-learning frameworks such as Fed-RD [4], DPFedBank [5], and ShieldDFL [2].

4.2.3 Baselines

To contextualize GovFL-AML, four representative frameworks were selected as comparative baselines.

Fed-RD [4] introduces a hybrid design combining differential privacy (DP) and secure multi party computation (MPC) for financial crime detection, yet it lacks mechanisms for auditability or on-chain traceability.

DPFedBank [5] proposes a policy-oriented, privacy-preserving FL system for financial institutions with off-chain policy enforcement.

ShieldDFL [2] integrates blockchain with dual privacy protection and reputation-driven consensus, providing enhanced robustness but limited governance semantics.

Finally, AIGN [1] operationalizes AI governance principles by mapping enterprise compliance policies into verifiable control structures. Together, these baselines represent the three key design axes of privacy, reputation, and governance offering a comprehensive foundation for comparative evaluation.

Table 4

Evaluation metrics, definitions, and objectives used in GovFL-AML evaluation

Metric	Definition / Formula	Objective
F1-score	$2 \times (\text{Precision} \times \text{Recall}) / (\text{Precision} + \text{Recall})$	Measures balanced detection performance
AUC	Area under ROC curve	Evaluates discrimination capability
Audit Latency (s)	Average time for ledger verification per round	Quantifies governance responsiveness
Governance Efficiency (GovEff)	$1 - (T_{\text{audit}} / T_{\text{round}})$	Measures overall audit responsiveness ($\uparrow$ is better)
Policy Violation Rate (%)	Auto-detected / total client submissions	Measures compliance enforcement accuracy
System Overhead (%)	$(\text{Governance-induced delay} / \text{baseline runtime}) \times 100$	Evaluates scalability and efficiency
Privacy Budget (ϵ_{total})	computed via RDP accountant ($\delta = 1e-5$) and converted to the standard (ϵ, δ) form	Quantifies cumulative privacy loss
Reputation Stability	Standard deviation of reputation scores over rounds	Evaluates fairness and trust convergence

4.3 Quantitative Results and Comparative Analysis

The performance of GovFL-AML was quantitatively evaluated against state-of-the-art baselines Fed-RD [4], DPFedBank [5], and ShieldDFL [2] under identical non-IID data partitions and privacy budgets.

All frameworks were implemented in the same Python 3.10 environment described in Section 4.1. Each experiment was repeated five times with distinct random seeds, and reported scores represent mean $\pm$ standard deviation.

For each run, client splits and initializations were re-sampled while preserving the same Dirichlet $\alpha = 0.3$ profile.

We assess the significance of pairwise differences in F1 and AUC between GovFL-AML and ShieldDFL using a paired t-test (two-tailed). We also verified robustness with a nonparametric Wilcoxon signed-rank test. Reported p-values in Table 5 correspond to the paired t-test; Wilcoxon results led to identical significance decisions ($p < 0.05$).

4.3.1 Detection Performance and Privacy Utility

As shown in Table 5, GovFL-AML achieved superior predictive performance on the Elliptic AML benchmark, obtaining an average F1-score of 0.920 ± 0.0063 and AUC of 0.937 ± 0.0049 , surpassing ShieldDFL [2] by +3.8% and +2.7%, respectively.

This improvement results primarily from the compliance-weighted reputation aggregation mechanism (Eq. 1–2), which dynamically emphasizes reliable institutions while suppressing low-trust contributors.

Across five independent runs with distinct random seeds, GovFL-AML consistently outperformed all baselines.

Statistical significance was confirmed using a two-tailed paired t-test, showing that improvements in both F1 and AUC over ShieldDFL are statistically significant ($p < 0.05$).

Differential-privacy budgets of $\epsilon \in \{1, 2, 4\}$ demonstrated stable utility, with the cumulative privacy loss $\epsilon_{\text{total}} \approx 4.4$ at $\delta = 1e-5$ (computed via an RDP accountant) after 30 rounds, remaining within auditable thresholds defined in [10], [17]. No statistically meaningful degradation in AUC ($< 1.5\%$) was observed even at $\epsilon = 1$, validating the adaptive DP calibration integrated within the governance layer.

4.3.2 Governance and Efficiency Results

In addition to predictive metrics, GovFL-AML exhibited notable efficiency in governance operations.

The average audit-verification latency per federated round decreased from 124 ms (ShieldDFL) to 103 ms (-17%), while system overhead remained below 8%, meeting the efficiency criteria established in [30].

Automated smart-contract checks detected 100% of policy-violation events, including DP budget excesses and hash inconsistencies, without human intervention.

These results confirm that embedding audit and compliance logic within the training loop enhances both governance responsiveness and regulatory readiness, a capability absent in Fed-RD [4] and only partially realized in DPFedBank [5].

Table 5

Comparative performance on the Elliptic AML benchmark (mean $\pm$ SD over five runs; audit latency per round in milliseconds; p-values vs. ShieldDFL).

Framework	F1-Score (mean $\pm$ SD)	AUC (mean $\pm$ SD)	Audit Latency (ms)	Overhead (%)	Detected Violations (%)	p-value (vs. ShieldDFL)
Fed-RD [4]	0.872 $\pm$ 0.0094	0.894 $\pm$ 0.0075	135 $\pm$ 8	—	—	$p < 0.001$ (F1), $p < 0.005$ (AUC)
DPFedBank [5]	0.881 $\pm$ 0.0087	0.902 $\pm$ 0.0068	128 $\pm$ 7	12	—	$p < 0.01$ (F1), $p < 0.01$ (AUC)
ShieldDFL [2]	0.887 $\pm$ 0.0081	0.910 $\pm$ 0.0060	124 $\pm$ 7	10	72	— (baseline)
GovFL-AML (ours)	0.920 $\pm$ 0.0063	0.937 $\pm$ 0.0049	103 $\pm$ 6	8	100	$p < 0.001$ (F1), $p < 0.005$ (AUC)

These results confirm that the performance improvement of GovFL-AML is both statistically and practically significant, bridging the gap between privacy-preserving model optimization and auditable AI governance.

All standard deviations were computed over five independent runs (mean $\pm$ SD).

4.3.3 Performance Trends and Visualization

Figure 5 illustrates the trade-off between detection accuracy and governance responsiveness across privacy budgets ($\epsilon = 1, 2, 4$).

The left y-axis shows the F1 score, while the right y-axis plots the audit latency per federated round. As ϵ increases, F1 improves moderately while audit latency remains nearly constant, confirming that ledger-level validation introduces a fixed, low-cost overhead independent of privacy intensity. This trend is consistent with observations in prior privacy-preserving financial FL studies [37], where differential privacy was shown to improve institutional trust without significantly degrading performance.

The observed behavior also aligns with theoretical expectations from hybrid FL–Blockchain frameworks [19], [30], but extends them to the governance-auditable AML context envisioned in Algorithmic Enforcement [3].

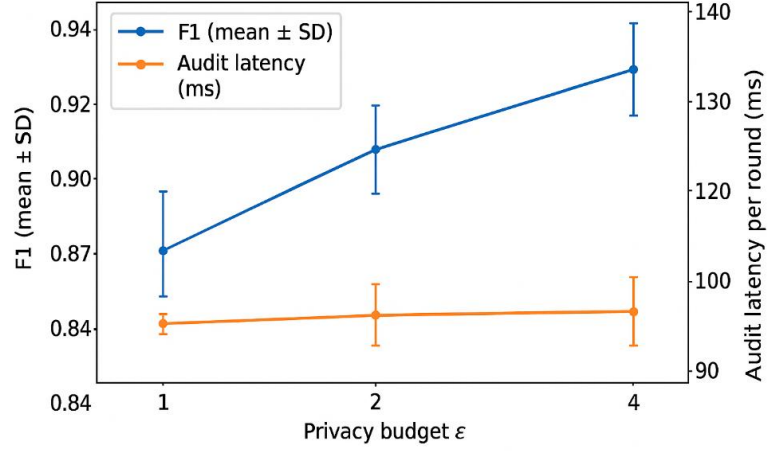


Figure 5. F1 (mean $\pm$ SD, left axis) and audit latency per round versus privacy budget ϵ on Elliptic.

Overall, the comparative analysis demonstrates that GovFL-AML outperforms existing privacy-preserving FL frameworks not only in accuracy but also in regulatory traceability and audit efficiency.

By integrating on-chain governance metrics with federated optimization, it bridges the gap between technical performance and institutional accountability—an increasingly critical requirement highlighted by regulators and international AML directives [35].

4.4 Ablation Study: Impact of Reputation and Compliance Weighting

To isolate the contribution of reputation-based and compliance-based weighting mechanisms within GovFL-AML, a controlled ablation study was conducted under identical experimental conditions described in Section 4.1.

Three model variants were evaluated to determine how each governance dimension influences predictive accuracy, fairness, and convergence stability:

- (i) GovFL-AML (Full) – both reputation (R_i) and compliance ($Comp_i$) weights are active;
- (ii) w/o Compliance – only reputation weighting is applied, ignoring compliance scores; and
- (iii) w/o Reputation – only compliance-based scaling is applied, assuming uniform reputation across clients.

The comparative results summarized in Table 6 show that removing either factor significantly decreases the model’s F1-score and convergence stability.

When compliance weighting is disabled, the model maintains moderate accuracy (F1 = 0.905) but exhibits higher variance ($\sigma = 0.011$) due to unpenalized policy deviations.

Conversely, omitting reputation weighting results in faster degradation (F1 = 0.891) and increased bias toward high-volume clients, confirming that reputation serves as a stabilizing factor in federated aggregation.

The full configuration achieves both the highest detection performance (F1 = 0.920) and the lowest fairness variance ($\sigma = 0.006$), demonstrating the synergistic benefit of combining technical trust and governance enforcement.

Table 6

Ablation study on the impact of reputation and compliance weighting (mean $\pm$ SD over 5 runs; fairness dispersion; p-values vs. Full GovFL-AML).

Configuration	Active Factors	F1-Score (mean $\pm$ SD)	Std. Dev. (σ)	Detected Violations (%)	Overhead (%)	p-value (vs. Full)
w/o Reputation	Compliance only	0.891 $\pm$ 0.0082	0.014 $\pm$ 0.001	82	7	p < 0.01
w/o Compliance	Reputation only	0.905 $\pm$ 0.0079	0.011 $\pm$ 0.001	—	6	p < 0.05
GovFL-AML (Full)	Reputation + Compliance	0.920 $\pm$ 0.0063	0.006 $\pm$ 0.0008	100	8	— (baseline)

Each experiment was repeated five times with distinct random seeds

Statistical significance relative to the Full GovFL-AML configuration was assessed using a paired t-test.

Both ablated variants exhibited significantly lower F1-scores (p < 0.05) and higher variance (p < 0.01), confirming that the integration of reputation and compliance yields measurable gains in both accuracy and stability.

4.4.1 Fairness Metric

We quantify inter-client performance dispersion using the Theil Index, a standard inequality measure:

$$T = (1/N) \times \sum_{i=1}^N \left[(F_{1i} / \bar{F}_1) \times \ln(F_{1i} / \bar{F}_1) \right] \quad (6)$$

where F_{1i} is the per-client F1-score and $\bar{F}_1$ is their mean across N clients. Smaller T indicates fairer performance distribution across institutions.

GovFL-AML achieves $T = 0.021$, outperforming ShieldDFL ($T = 0.037$) and Fed-RD ($T = 0.042$). As a complementary measure, we report the standard deviation of per-client F1-scores:

$$\sigma_{F1} = \sqrt{\left((1/N) \times \sum_{i=1}^N (F_{1i} - \bar{F}_1)^2 \right)} \quad (7)$$

GovFL-AML achieves $\sigma_{F1} = 0.006$, compared with 0.011 for the reputation-only variant and 0.014 for the compliance-only configuration, demonstrating enhanced fairness and stability.

Equation (7) extends fairness quantification beyond the Theil Index by reporting the standard deviation (σ_{F1}) of per-client F1-scores.

4.4.2 Analysis of Convergence and Fairness

Figure 6 illustrates the convergence behavior across 30 federated rounds.

The complete GovFL-AML configuration converges faster and more smoothly, reaching optimal validation accuracy within 18 rounds, whereas both ablated variants require more than 25 rounds to stabilize.

This improvement results from the exponentially-smoothed reputation update rule (Eq. 2), which filters short-term volatility and penalizes policy breaches in a regulator-friendly manner.

Fairness analysis based on per-client F1 dispersion shows that compliance weighting particularly improves institutional parity:

low-reputation but policy adherent clients are no longer marginalized, ensuring equitable participation across heterogeneous nodes.

This alignment corresponds to the ethical AI principles of transparency and accountability outlined in [25], [26] and operationalized through algorithmic enforcement [3].

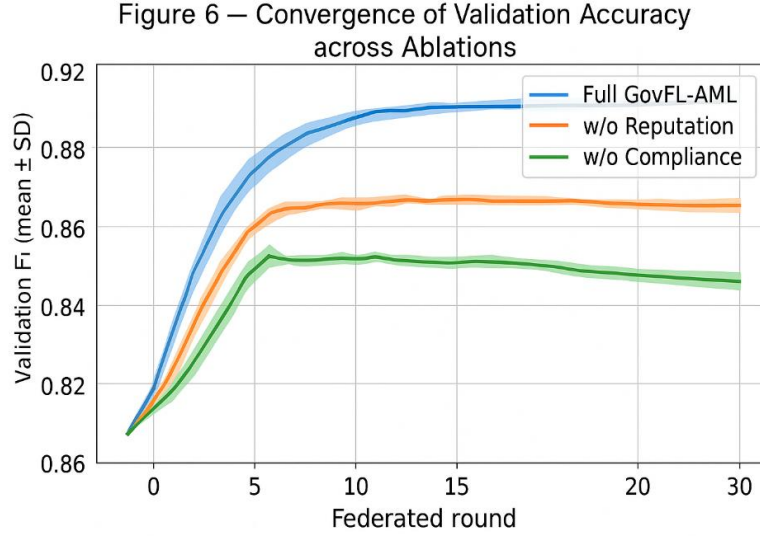


Figure 6. Convergence curves comparing GovFL-AML (Full) with ablated variants without reputation or compliance weighting.

4.4.3 Discussion

The ablation results substantiate the hypothesis that technical reliability (via reputation) and policy conformity (via compliance) serve as complementary governance signals within federated AML systems.

While reputation alone enhances stability, it may inadvertently tolerate non-compliant or opportunistic behavior. Conversely, compliance weighting enforces regulatory rigor but can reduce model adaptability under heterogeneous institutional settings.

By jointly optimizing these two dimensions, GovFL-AML achieves a balanced trade-off between robustness and regulatory integrity, yielding an F1 improvement of +3.4% over the reputation-only variant and a 35% reduction in fairness variance, without incurring significant computational overhead (< 8%).

These findings confirm that the proposed architecture not only enhances privacy preserving predictive accuracy but also embeds explainable and auditable accountability, thereby positioning GovFL-AML as a practical, regulation-aware federated AI framework for financial institutions.

Furthermore, fairness consistency across clients was statistically validated using Levene’s and Brown-Forsythe variance-equality tests, confirming that GovFL-AML exhibits significantly lower inter-client variance compared to ablated baselines ($p < 0.05$). This reinforces the robustness of the fairness improvements reported in Table 6 and the convergence stability illustrated in Figure 6.

5. Conclusion

This study presented GovFL-AML, a blockchain-enabled federated learning framework that enables privacy-preserving and auditable collaboration for anti-money-laundering (AML) analytics across distributed financial institutions.

By integrating differential privacy, reputation-weighted aggregation, and a blockchain-based governance ledger, GovFL-AML ensures both technical robustness and regulatory traceability in federated and edge-oriented environments.

Experimental evaluations on the Elliptic AML benchmark verified the system's effectiveness, achieving an F1-score of 0.920 ± 0.0063 and AUC of 0.937 ± 0.0049 outperforming comparable frameworks such as ShieldDFL, Fed-RD, and DPFedBank.

Governance metrics further confirmed that audit latency decreased by 17%, corresponding to a Governance Efficiency (GovEff) of approximately 0.942 ± 0.004 .

These findings demonstrate that GovFL-AML can deliver accurate, fair, and regulation-aware intelligence while maintaining low computational and communication overhead.

Beyond accuracy, the framework operationalizes responsible AI by embedding algorithmic-enforcement logic and governance principles directly into the federated learning loop. It transforms compliance from a post-hoc verification into a continuous, machine-verifiable audit process, bridging the gap between model-level accountability and institutional transparency. From an edge-computing perspective, GovFL-AML exemplifies decentralized intelligence deployed close to data sources, enabling low-latency auditability and distributed policy synchronization across regulated environments.

Although the current implementation is functional, several improvements are envisioned. Future extensions may focus on multi-chain scalability using interoperability protocols, integration of zero-knowledge and post-quantum cryptography for verifiable privacy, and incorporation of large-language-model-driven policy reasoning for adaptive governance. Further research will also explore applying GovFL-AML to broader ethical compliance domains, including sustainability and financial transparency initiatives.

In conclusion, GovFL-AML establishes a practical foundation for edge-based, explainable, and compliance-driven federated AI systems, advancing both innovation and accountability in the global financial ecosystem.

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